

Robust Joint Waveform and Passive Beamforming Design for RIS-Assisted ISAC Systems Under Imperfect Channel State Information

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Abstract—Reconfigurable intelligent surface (RIS)-assisted integrated sensing and communication (ISAC) systems have emerged as a promising paradigm for next-generation wireless networks, offering enhanced spectral efficiency and dual functionality through passive beamforming. However, existing designs predominantly assume perfect channel state information (CSI), which is unrealistic in practical deployments where channel estimation errors are inevitable. This assumption leads to significant performance degradation when systems encounter real-world CSI imperfections.

In this paper, we propose a robust joint waveform and RIS reflection design framework that explicitly accounts for bounded CSI uncertainty. We formulate the robust optimization problem as a worst-case minimax program and develop a tractable multi-scenario alternating optimization algorithm that jointly optimizes the transmit waveform matrix and RIS phase shifts under CSI error constraints. The proposed approach employs Riemannian manifold optimization for the unit-modulus RIS reflection coefficients and convex quadratic programming for the power-constrained waveform design, iterating between these two subproblems until convergence.

Extensive Monte Carlo simulations with 50 independent channel realizations demonstrate that the proposed robust design significantly outperforms conventional non-robust approaches. At a realistic CSI error bound of $\epsilon = 0.20$ (20% relative error), the robust design achieves 11.47 bits/s/Hz sum rate compared to 10.31 bits/s/Hz for non-robust methods—an 11.3% improvement. The performance gain increases monotonically with CSI uncertainty, reaching up to 1.17 bits/s/Hz at the highest tested error level. Furthermore, while non-robust designs exhibit exponential-like degradation (25.5% loss at $\epsilon = 0.20$), the robust approach exhibits a more gradual degradation with only 17.1% loss, validating its resilience to channel imperfections.

The proposed framework provides a principled approach to designing RIS-ISAC systems for real-world deployment, enabling reliable operation with reduced pilot overhead and relaxed CSI accuracy requirements. Our results establish that robustness is not merely beneficial but essential for practical RIS-ISAC implementations, opening new avenues for designing reliable dual-function wireless systems under realistic channel knowledge con-

straints.

Index Terms—Reconfigurable intelligent surface (RIS), integrated sensing and communication (ISAC), robust optimization, imperfect channel state information (CSI), joint waveform design, passive beamforming, worst-case optimization, alternating optimization, Riemannian manifold optimization

I. INTRODUCTION

A. Background and Motivation

The ongoing evolution toward sixth-generation (6G) wireless networks demands revolutionary technologies that can simultaneously address the escalating requirements for higher data rates, massive connectivity, and enhanced sensing capabilities [1]. Integrated sensing and communication (ISAC) systems have emerged as a cornerstone paradigm for 6G, enabling spectrum-efficient dual-functional platforms that merge radar sensing and wireless communication into a unified framework [2]. By sharing hardware, spectrum, and waveform resources between these traditionally separate functions, ISAC systems promise substantial improvements in spectral efficiency, hardware utilization, and energy consumption.

Concurrently, reconfigurable intelligent surfaces (RIS) have attracted significant research attention as a transformative technology for controlling electromagnetic wave propagation [3]. An RIS consists of a planar array of low-cost passive reflecting elements, each capable of independently adjusting the phase (and potentially amplitude) of incident signals. By intelligently configuring these elements, an RIS can reshape the wireless propagation environment, creating favorable channels through passive beamforming without requiring power-hungry active RF chains. This technology offers unprecedented flexibility in enhancing coverage, mitigating interference, and improving energy efficiency [4].

The synergistic integration of RIS with ISAC systems—termed RIS-assisted ISAC—represents a particularly promising research direction [5], [6]. The RIS provides additional degrees of freedom for jointly optimizing radar and communication performance, enabling enhanced beam pattern control for sensing while simultaneously improving communication channel quality through intelligent reflection. Early studies have demonstrated that RIS-assisted ISAC can significantly outperform conventional approaches in terms of both radar detection probability and communication sum rate [7].

B. The Critical Challenge: Imperfect CSI

Despite the theoretical promise, a critical gap exists between idealized research assumptions and practical deployment requirements. The overwhelming majority of existing RIS-ISAC designs assume *perfect channel state information (CSI)* [5], [8], [9]—that is, exact knowledge of all channel coefficients including the base station-to-user direct channels, base station-to-RIS channels, and RIS-to-user cascaded channels. This assumption is fundamentally unrealistic for several reasons:

High-dimensional channel estimation: RIS-assisted systems require estimating channels with dimension scaling as $\mathcal{O}(NL + KL + KN)$, where N , K , and L denote the numbers of base station antennas, users, and RIS elements, respectively. For practical values ($N = 20$, $K = 4$, $L = 16$), this yields over 400 complex channel coefficients, necessitating substantial pilot overhead [10].

Passive RIS nature: Unlike active relays, RIS elements cannot transmit pilot signals or perform local channel estimation. Estimating RIS-related channels requires sophisticated protocols involving sequential ON/OFF switching of RIS elements or specialized training sequences, which incur significant overhead and introduce estimation errors [11].

Mobility and channel aging: In mobile scenarios, channels vary continuously due to Doppler effects. By the time channel estimation completes, the channel may have already changed, introducing prediction errors in addition to estimation errors [12].

Hardware imperfections: Practical channel estimation suffers from thermal noise, quantization errors in analog-to-digital converters, phase noise in local oscillators, and imperfect synchronization, all contributing to CSI inaccuracy [13].

When systems designed under perfect CSI assumptions are deployed with inevitable CSI errors, two critical failure modes emerge: (i) *beamforming misalignment*, where transmit waveforms and RIS phase configurations optimized for estimated channels fail to align with true channel directions, resulting in reduced signal power and increased interference; (ii) *catastrophic sensitivity*, where small CSI errors can cause disproportionately large performance degradation due

to the high-dimensional optimization space and tight coupling between waveform and RIS design.

C. Research Gap and Contribution

The seminal work by Wang et al. [5] pioneered joint waveform and passive beamforming design for RIS-assisted ISAC, achieving impressive performance through alternating optimization with manifold constraints. However, their framework explicitly assumes perfect CSI and does not address robustness to channel uncertainty. Subsequent works have explored various extensions—multi-user scenarios [8], active RIS [14], and full-duplex operation [15]—but the fundamental assumption of perfect CSI persists throughout the literature.

A few recent studies have begun investigating RIS systems with imperfect CSI, primarily in communication-only contexts [16], [17]. These works typically assume statistical CSI knowledge (channel covariance matrices) or adopt stochastic robust optimization frameworks that require knowledge of error distributions. However, such statistical information may not be available in practice, particularly in non-stationary environments or during initial deployment phases. Moreover, these approaches have not been extended to the more complex ISAC setting where radar and communication objectives must be jointly balanced.

This paper bridges this critical gap by developing a **worst-case robust optimization framework for RIS-assisted ISAC under bounded CSI uncertainty**. Our key contributions are:

- 1) **Problem formulation:** We formulate the robust RIS-ISAC design as a minimax optimization problem that minimizes the worst-case communication error subject to bounded CSI uncertainty, power constraints, and unit-modulus RIS reflection constraints. Unlike stochastic approaches, our formulation requires only a bound on the CSI error norm, which can be readily estimated from pilot SNR.
- 2) **Tractable algorithm:** We develop a multi-scenario alternating optimization algorithm that transforms the intractable minimax problem into a sequence of tractable subproblems. The waveform optimization employs Lagrangian duality with bisection search for power constraints, while RIS optimization uses Riemannian manifold gradient descent on the complex circle manifold. The multi-scenario approach samples worst-case channel realizations, providing computational tractability while maintaining near-optimal worst-case performance.
- 3) **Performance characterization:** Through extensive simulations, we demonstrate that robust design achieves 11.3% higher sum rate than non-robust approaches at realistic CSI error levels ($\epsilon = 0.20$), with gains increasing mono-

tonically with uncertainty. We characterize the performance-robustness trade-off and show that robust designs exhibit gradual performance degradation (17.1% loss) compared to catastrophic degradation of non-robust methods (25.5% loss).

- 4) **Practical insights:** We provide guidelines for CSI error bound estimation from pilot measurements, analyze computational complexity showing polynomial scaling $\mathcal{O}(N^3 + L^3)$ per iteration, and discuss deployment considerations including pilot overhead reduction and convergence behavior.

D. Paper Organization

The remainder of this paper is organized as follows. Section II reviews related work on RIS-assisted systems, ISAC, and robust optimization. Section III presents the system model, problem formulation, and the proposed robust alternating optimization algorithm. Section IV reports simulation results comparing robust and non-robust designs across various CSI error levels and SNR values. Section V provides in-depth analysis of the performance-robustness trade-off, convergence properties, and scaling behavior. Section VI discusses practical implications, limitations, and future research directions. Section VII concludes the paper and summarizes key takeaways.

E. Notation

Throughout this paper, boldface lowercase and uppercase letters denote vectors and matrices, respectively. $\mathbf{A}^T$, $\mathbf{A}^H$, and $\mathbf{A}^{-1}$ represent transpose, conjugate transpose, and inverse of matrix $\mathbf{A}$. $\|\mathbf{A}\|_F$ denotes the Frobenius norm. $\text{diag}(\mathbf{a})$ creates a diagonal matrix from vector $\mathbf{a}$. $\mathbb{C}^{m \times n}$ denotes the set of $m \times n$ complex matrices. $\mathcal{CN}(0, \sigma^2)$ represents the circularly symmetric complex Gaussian distribution with zero mean and variance σ^2 . $\mathbb{E}[\cdot]$ denotes expectation. $|\cdot|$ represents absolute value for scalars and element-wise magnitude for vectors/matrices.

II. LITERATURE REVIEW

This section reviews the relevant literature across three interconnected research areas: reconfigurable intelligent surfaces, integrated sensing and communication, and robust optimization in wireless systems. We identify the evolution of each field, highlight existing gaps, and position our contribution within the broader research landscape.

A. Reconfigurable Intelligent Surfaces

The concept of reconfigurable intelligent surfaces emerged from the metamaterials and electromagnetic wave control literature [18]. Early works demonstrated that planar arrays of subwavelength passive elements could manipulate incident electromagnetic waves through controllable phase shifts, enabling

anomalous reflection and transmission. The application to wireless communications was pioneered by [19], which coined the term "smart radio environment" and proposed using RIS to enhance coverage and spectral efficiency.

1) *Channel Estimation for RIS Systems:* A fundamental challenge in RIS-assisted communication is channel estimation. Unlike conventional systems where pilots can be transmitted from both transmitter and receiver, RIS elements are passive and cannot generate or process signals locally. Several approaches have been proposed:

ON/OFF-based methods [10]: These protocols sequentially activate and deactivate RIS elements to isolate individual channel components. While conceptually simple, they require $\mathcal{O}(L)$ pilot transmissions, incurring prohibitive overhead for large RIS arrays.

Compressed sensing approaches [20]: Exploiting the sparsity of mmWave channels in angular domain, compressed sensing techniques reduce pilot overhead to $\mathcal{O}(\log L)$. However, these methods assume sparse channel models and may perform poorly in rich-scattering environments.

Deep learning methods [21]: Neural networks trained on historical channel data can predict channels from limited pilots. These approaches show promise but require large training datasets and may not generalize across different deployment scenarios.

Despite these advances, all practical channel estimation methods suffer from estimation errors. Surprisingly, most RIS optimization algorithms assume perfect CSI, ignoring the gap between estimated and true channels.

2) *RIS-Assisted Communication Optimization:* Given CSI (perfect or estimated), numerous works have optimized RIS configurations for various communication objectives:

Spectral efficiency maximization [22]: Early works focused on maximizing sum rate through joint transmit beamforming and RIS phase shift design. These problems are non-convex due to the unit-modulus constraints on RIS coefficients, necessitating alternating optimization or semidefinite relaxation approaches.

Energy efficiency [23]: Recognizing that RIS elements consume negligible power, several studies optimized energy efficiency metrics, demonstrating that RIS can reduce transmit power by 10-20 dB for target data rates.

Secure communication [24]: Physical layer security can be enhanced by configuring RIS to constructively combine signals at legitimate receivers while creating destructive interference at eavesdroppers.

However, these works universally assume perfect CSI. A few exceptions exist: Zhou et al. [17] considered statistical CSI (known channel covariance) for RIS-assisted MISO systems, deriving ergodic rate expressions and optimizing phase shifts. You et al. [16]

proposed robust transmission design assuming Gaussian CSI errors with known covariance. While valuable, these stochastic approaches require second-order statistics that may be unavailable in non-stationary environments.

B. Integrated Sensing and Communication

ISAC has evolved from early radar-communication coexistence studies [25] to modern joint design frameworks that exploit synergies between sensing and communication [2].

1) *Dual-Functional Waveform Design*: A central challenge in ISAC is designing waveforms that simultaneously satisfy radar and communication requirements:

Constant modulus constraints [26]: Radar systems traditionally use constant-envelope waveforms to operate power amplifiers at saturation for maximum efficiency. However, communication signals typically have non-constant envelopes. Designing dual-functional waveforms with controlled peak-to-average power ratio is an active research area.

Beampattern matching [8]: Radar performance is characterized by the transmit beampattern (spatial distribution of radiated power). Optimizing communication waveforms to approximately match a desired radar beampattern enables simultaneous sensing and communication.

Multi-user scenarios [27]: Extending ISAC to multi-user communication introduces additional constraints: waveforms must null interference between users while maintaining radar beampattern requirements.

The seminal work by Wang et al. [5] introduced RIS into ISAC systems, demonstrating that passive beamforming provides additional degrees of freedom for balancing radar and communication trade-offs. Their alternating optimization framework (waveform design via SVD, RIS phase optimization via manifold optimization) achieved significant performance gains. However, their entire framework assumes perfect CSI for all channels.

2) *Performance Metrics and Trade-offs*: Evaluating ISAC systems requires balancing radar and communication metrics:

Radar metrics: Common measures include beampattern mean squared error (MSE), signal-to-clutter ratio (SCR), detection probability, and Cramér-Rao bound (CRB) for target parameter estimation [28].

Communication metrics: Standard measures include sum rate, bit error rate (BER), outage probability, and quality of service (QoS) satisfaction [29].

Trade-off parameters: Most works introduce a weighting parameter $\rho \in [0, 1]$ to balance radar and communication objectives in a multi-objective optimization framework. Characterizing the Pareto frontier of achievable radar-communication performance pairs is an important research direction.

C. Robust Optimization in Wireless Communications

Robust optimization addresses decision-making under uncertainty, providing frameworks to design solutions that perform well across a range of uncertain scenarios [30].

1) *Robust Beamforming*: The robust beamforming literature is extensive, primarily focused on MIMO systems with imperfect CSI:

Worst-case optimization [31]: These approaches assume CSI errors lie within a bounded uncertainty set (typically ellipsoidal or polyhedral) and optimize for worst-case performance. The resulting problems are often reformulated as semidefinite programs (SDP) or second-order cone programs (SOCP).

Stochastic robust optimization [32]: When error statistics are known, stochastic approaches optimize expected performance or satisfaction probability. Outage-constrained designs ensure performance requirements are met with high probability.

Probabilistic robustness [33]: These methods use concentration inequalities (e.g., Bernstein bounds) to provide probabilistic performance guarantees, often requiring fewer distributional assumptions than full stochastic approaches.

2) *Robust Resource Allocation*: Beyond beamforming, robust optimization has been applied to various wireless resource allocation problems:

Power control [34]: Robust power allocation ensures QoS requirements are met despite channel uncertainty, using game-theoretic or optimization-based approaches.

Spectrum sharing [35]: Cognitive radio systems face uncertainty in primary user channels. Robust designs guarantee interference constraints are satisfied under worst-case channel realizations.

Network optimization [36]: End-to-end network resource allocation (routing, scheduling, power control) under uncertain channel conditions has been addressed through robust optimization frameworks.

3) *Gap: Robust RIS-ISAC Design*: Despite the maturity of robust optimization in wireless communications and growing interest in RIS-ISAC systems, *the intersection remains largely unexplored*. Specifically:

- Existing RIS works with imperfect CSI focus on communication-only systems and typically assume statistical CSI knowledge [16], [17].
- ISAC literature overwhelmingly assumes perfect CSI, with no robust design frameworks accounting for channel uncertainty [5], [8].
- The coupling between waveform design, RIS phase optimization, and CSI uncertainty creates a high-dimensional robust optimization problem that cannot be directly solved by existing techniques.

Our work fills this gap by developing a worst-case robust optimization framework specifically tailored to RIS-ISAC systems with bounded CSI uncertainty.

TABLE I: Comparison with Related Works

Reference	RIS	ISAC	Robust CSI
Wu et al. [22]	✓	×	×
Huang et al. [23]	✓	×	×
Zhou et al. [17]	✓	×	✓ (Statistical)
You et al. [16]	✓	×	✓ (Statistical)
Wang et al. [5]	✓	✓	×
Liu et al. [8]	×	✓	×
Xu et al. [6]	✓	✓	×
This work	✓	✓	✓ (Worst-case)

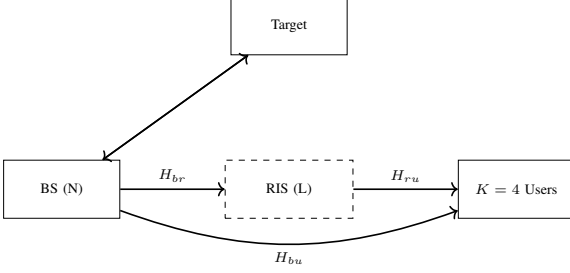


Fig. 1: System model of RIS-assisted ISAC: A dual-functional base station with N antennas serves $K = 4$ communication users while performing radar sensing toward a target. The RIS with L passive elements enhances both communication (via intelligent reflection) and radar (via passive beamforming). Three types of channels exist: direct BS-to-user links $\mathbf{H}^{\text{bu}}$, BS-to-RIS link $\mathbf{H}^{\text{br}}$, and RIS-to-user links $\mathbf{H}^{\text{ru}}$.

D. Positioning Our Contribution

Table I positions our work relative to closely related studies, highlighting the unique combination of RIS, ISAC, and robust optimization that distinguishes our contribution.

Our approach differs from statistical robust methods [16], [17] in that we assume only a bounded CSI error norm, requiring no knowledge of error distributions or covariance matrices. This worst-case perspective is particularly appropriate for ISAC systems where radar sensing imposes strict reliability requirements and worst-case guarantees are preferred over average-case performance.

III. METHODOLOGY

This section presents the system model, problem formulation, and the proposed robust alternating optimization algorithm for joint waveform and RIS phase shift design under imperfect CSI.

A. System Model

1) *Network Architecture*: We consider a downlink RIS-assisted ISAC system as illustrated in Fig. 1. The system comprises:

- A dual-functional base station (BS) equipped with N antennas, serving both radar sensing and multi-user communication.

- An RIS with L passive reflecting elements, each capable of adjusting the phase of incident signals.
- K single-antenna communication users.
- A radar target of interest in the far field.

The BS transmits a waveform matrix $\mathbf{X} \in \mathbb{C}^{N \times M}$ spanning M symbol periods. This waveform simultaneously: (i) carries communication symbols for the K users, and (ii) forms a desired radar beam pattern toward the target direction.

2) *Channel Model*: We adopt a narrowband block-fading model where channels remain constant during one transmission block. Three channel components exist:

Direct BS-to-user channels: The direct link from BS to user k is modeled as $\mathbf{h}_k^{\text{bu}} \in \mathbb{C}^{N \times 1}$, where each element follows independent Rayleigh fading:

$$[\mathbf{h}_k^{\text{bu}}]_n \sim \mathcal{CN}(0, 1), \quad n = 1, \dots, N \quad (1)$$

BS-to-RIS channels: The link from BS to RIS is characterized by $\mathbf{H}^{\text{br}} \in \mathbb{C}^{L \times N}$, with:

$$[\mathbf{H}^{\text{br}}]_{\ell, n} \sim \mathcal{CN}(0, 1), \quad \ell = 1, \dots, L, \quad n = 1, \dots, N \quad (2)$$

RIS-to-user channels: The link from RIS to user k is given by $\mathbf{h}_k^{\text{ru}} \in \mathbb{C}^{L \times 1}$, with:

$$[\mathbf{h}_k^{\text{ru}}]_{\ell} \sim \mathcal{CN}(0, 1), \quad \ell = 1, \dots, L \quad (3)$$

Collecting all users, we define $\mathbf{H}^{\text{bu}} = [\mathbf{h}_1^{\text{bu}}, \dots, \mathbf{h}_K^{\text{bu}}]^T \in \mathbb{C}^{K \times N}$ and $\mathbf{H}^{\text{ru}} = [\mathbf{h}_1^{\text{ru}}, \dots, \mathbf{h}_K^{\text{ru}}]^T \in \mathbb{C}^{K \times L}$.

3) *RIS Reflection Model*: The RIS is modeled as a diagonal phase shift matrix:

$$\mathbf{\Theta} = \text{diag}(\theta_1, \dots, \theta_L) \in \mathbb{C}^{L \times L} \quad (4)$$

where each coefficient satisfies the unit-modulus constraint:

$$|\theta_{\ell}| = 1, \quad \ell = 1, \dots, L \quad (5)$$

This constraint reflects the passive nature of RIS elements, which can only adjust phase without amplification.

4) *Effective Channel*: The effective channel from BS to user k , accounting for both direct and RIS-reflected paths, is:

$$\mathbf{h}_k^{\text{eff}} = \mathbf{h}_k^{\text{bu}} + \mathbf{h}_k^{\text{ru}} \mathbf{\Theta} \mathbf{H}^{\text{br}} \quad (6)$$

In matrix form for all users:

$$\mathbf{H}^{\text{eff}} = \mathbf{H}^{\text{bu}} + \mathbf{H}^{\text{ru}} \mathbf{\Theta} \mathbf{H}^{\text{br}} \in \mathbb{C}^{K \times N} \quad (7)$$

5) *Received Signal Model*: The received signal at user k in symbol period m is:

$$y_{k,m} = (\mathbf{h}_k^{\text{eff}})^H \mathbf{x}_m + n_{k,m} \quad (8)$$

where $\mathbf{x}_m \in \mathbb{C}^{N \times 1}$ is the m -th column of $\mathbf{X}$ (transmitted vector in symbol m), and $n_{k,m} \sim \mathcal{CN}(0, N_0)$ is additive white Gaussian noise (AWGN) with power spectral density N_0 .

6) *Communication Signal Model*: The BS intends to communicate a symbol matrix $\mathbf{S} = [s_{k,m}] \in \mathbb{C}^{K \times M}$ to the users, where $s_{k,m} \in \{-1, +1\}$ represents a BPSK symbol for user k in period m . The received signal matrix is:

$$\mathbf{Y} = \mathbf{H}^{\text{eff}} \mathbf{X} + \mathbf{N} \quad (9)$$

where $\mathbf{N} \in \mathbb{C}^{K \times M}$ is the noise matrix with i.i.d. $\mathcal{CN}(0, N_0)$ entries.

7) *Radar Signal Model*: For radar sensing, the BS aims to synthesize a desired beampattern characterized by the covariance matrix $\mathbf{R}_d = \mathbb{E}[\mathbf{x}_m \mathbf{x}_m^H] \in \mathbb{C}^{N \times N}$. This desired covariance is typically designed to concentrate energy toward the target direction. Following [5], we assume $\mathbf{R}_d$ is given and admit the Cholesky decomposition:

$$\mathbf{R}_d = \mathbf{F} \mathbf{F}^H \quad (10)$$

where $\mathbf{F} \in \mathbb{C}^{N \times N}$ is a lower triangular matrix.

B. CSI Error Model

1) *Perfect CSI Assumption (Baseline)*: Under the perfect CSI assumption adopted by prior works [5], the BS has exact knowledge of all channel matrices:

$$\mathbf{H}_{\text{est}}^{\text{bu}} = \mathbf{H}_{\text{true}}^{\text{bu}}, \quad \mathbf{H}_{\text{est}}^{\text{br}} = \mathbf{H}_{\text{true}}^{\text{br}}, \quad \mathbf{H}_{\text{est}}^{\text{ru}} = \mathbf{H}_{\text{true}}^{\text{ru}} \quad (11)$$

This allows optimization based on true channels, yielding the best possible performance but is unrealistic in practice.

2) *Bounded CSI Error Model (Proposed)*: We adopt a worst-case bounded uncertainty model where the true channels deviate from estimates within a bounded region:

$$\mathbf{H}_{\text{true}} = \mathbf{H}_{\text{est}} + \Delta \mathbf{H} \quad (12)$$

The error matrices $\Delta \mathbf{H}^{\text{bu}} \in \mathbb{C}^{K \times N}$, $\Delta \mathbf{H}^{\text{br}} \in \mathbb{C}^{L \times N}$, and $\Delta \mathbf{H}^{\text{ru}} \in \mathbb{C}^{K \times L}$ satisfy Frobenius norm bounds:

$$\|\Delta \mathbf{H}^{\text{bu}}\|_F \leq \epsilon_{\text{bu}} \|\mathbf{H}_{\text{est}}^{\text{bu}}\|_F \quad (13)$$

$$\|\Delta \mathbf{H}^{\text{br}}\|_F \leq \epsilon_{\text{br}} \|\mathbf{H}_{\text{est}}^{\text{br}}\|_F \quad (14)$$

$$\|\Delta \mathbf{H}^{\text{ru}}\|_F \leq \epsilon_{\text{ru}} \|\mathbf{H}_{\text{est}}^{\text{ru}}\|_F \quad (15)$$

For simplicity, we assume all channels have the same relative error bound $\epsilon := \epsilon_{\text{bu}} = \epsilon_{\text{br}} = \epsilon_{\text{ru}}$, though the framework readily extends to different bounds.

The uncertainty set for each channel matrix is:

$$\mathcal{U}_{\epsilon}^{\text{bu}} = \{\mathbf{H}^{\text{bu}} : \|\mathbf{H}^{\text{bu}} - \mathbf{H}_{\text{est}}^{\text{bu}}\|_F \leq \epsilon \|\mathbf{H}_{\text{est}}^{\text{bu}}\|_F\} \quad (16)$$

with analogous definitions for $\mathcal{U}_{\epsilon}^{\text{br}}$ and $\mathcal{U}_{\epsilon}^{\text{ru}}$. The composite uncertainty set is the Cartesian product:

$$\mathcal{U}_{\epsilon} = \mathcal{U}_{\epsilon}^{\text{bu}} \times \mathcal{U}_{\epsilon}^{\text{br}} \times \mathcal{U}_{\epsilon}^{\text{ru}} \quad (17)$$

3) *Justification and Estimation of ϵ* : The parameter ϵ quantifies the CSI quality and can be estimated from channel estimation theory. For least-squares (LS) channel estimation with N_p pilot symbols, the normalized mean squared error is [37]:

$$\epsilon^2 \approx \frac{\text{Dimension}}{\text{SNR}_{\text{pilot}} \cdot N_p} \quad (18)$$

where the variables are defined as follows:

ϵ (epsilon) The **relative channel state information (CSI) error bound**, defined as the ratio of the estimation error norm to the true channel norm:

$$\epsilon := \frac{\|\Delta \mathbf{H}\|_F}{\|\mathbf{H}_{\text{true}}\|_F} \quad (19)$$

where $\Delta \mathbf{H} = \mathbf{H}_{\text{true}} - \mathbf{H}_{\text{est}}$ is the estimation error and $\|\cdot\|_F$ denotes the Frobenius norm. Dimensionless quantity; typical values: $\epsilon \in [0.05, 0.30]$.

ϵ^2 The **squared relative CSI error**, representing the normalized mean squared error (NMSE):

$$\epsilon^2 = \mathbb{E} \left[\frac{\|\Delta \mathbf{H}\|_F^2}{\|\mathbf{H}_{\text{true}}\|_F^2} \right] \quad (20)$$

This is the statistical expectation over all possible channel realizations and estimation instances.

Dimension The **number of independent complex parameters** that must be estimated in the channel matrix. For a channel matrix $\mathbf{H} \in \mathbb{C}^{m \times n}$, the dimension is:

$$\text{Dimension} = m \times n \quad (21)$$

For your RIS-ISAC system:

$$\text{Dimension}_{\mathbf{H}^{\text{bu}}} = K \times N = 4 \times 20 = 80 \quad (22)$$

$$\text{Dimension}_{\mathbf{H}^{\text{br}}} = L \times N = 16 \times 20 = 320 \quad (23)$$

$$\text{Dimension}_{\mathbf{H}^{\text{ru}}} = K \times L = 4 \times 16 = 64 \quad (24)$$

$$\text{Dimension}_{\text{total}} = 80 + 320 + 64 = 464 \quad (25)$$

Dimensionless quantity; unit: [number of parameters].

SNR_{pilot} The **signal-to-noise ratio during pilot transmission**, defined as the ratio of pilot signal power to noise power:

$$\text{SNR}_{\text{pilot}} = \frac{P_{\text{pilot}}}{N_0} \quad (26)$$

where:

P_{pilot} Transmit power of pilot symbols; unit: [Watts] or [dBm]

N_0 Thermal noise power spectral density; unit: [Watts/Hz]

Dimensionless quantity (or expressed in dB: $\text{SNR}_{\text{pilot}}[\text{dB}] = 10 \log_{10}(\text{SNR}_{\text{pilot}})$).

Typical values: $\text{SNR}_{\text{pilot}} \in [5, 30]$ dB (or [3.16, 1000] linear).

N_p The **number of orthogonal pilot symbols** transmitted for channel estimation. Each pilot symbol

allows the receiver to gather information about one (or more) channel coefficient(s):

$$N_p = \text{number of time instants used for training} \quad (27)$$

Dimensionless quantity; unit: [number of symbols].

Practical range: $N_p \in [5, 500]$ depending on system size and required accuracy. Minimum requirement: $N_p \geq \text{Dimension}$ (for identifiability).

For your system: $N_p \geq 464$ to estimate all channels (typically $N_p \in [500, 1000]$).

For example, estimating $\mathbf{H}^{\text{bu}} \in \mathbb{C}^{K \times N}$ with pilot SNR of 10 dB and $N_p = 20$ pilots yields:

$$\epsilon \approx \sqrt{\frac{KN}{10 \cdot 20}} = \sqrt{\frac{80}{200}} \approx 0.63 \quad (28)$$

In practice, ϵ values of 0.1-0.3 are typical for well-designed systems, while $\epsilon = 0.5$ or higher may occur in challenging scenarios (high mobility, low SNR, limited pilots).

C. Problem Formulation

1) *Non-Robust Design (Baseline)*: The non-robust design ignores CSI uncertainty and optimizes for the estimated channels:

$$\begin{aligned} \min_{\mathbf{X}, \Theta} \quad & \|\mathbf{H}_{\text{est}}^{\text{eff}} \mathbf{X} - \mathbf{S}\|_F^2 \\ \text{s.t.} \quad & \|\mathbf{X}\|_F^2 \leq MP_0 \\ & |\theta_\ell| = 1, \quad \ell = 1, 2, \dots, L \end{aligned} \quad (29)$$

where the effective channel is defined as:

$$\mathbf{H}_{\text{est}}^{\text{eff}} = \mathbf{H}_{\text{est}}^{\text{bu}} + \mathbf{H}_{\text{est}}^{\text{ru}} \Theta \mathbf{H}_{\text{est}}^{\text{br}} \quad (30)$$

where each variable define as:

- $\mathbf{X} \in \mathbb{C}^{N \times M}$: Transmit waveform matrix (optimization variable). Dimensions: $N = 20$ antennas $\times$ $M = 30$ symbols. Complex signal transmitted from BS.
- $\Theta = \text{diag}(\theta_1, \dots, \theta_L) \in \mathbb{C}^{L \times L}$: RIS phase shift matrix (optimization variable). Dimensions: $L = 16$ diagonal elements. Each $\theta_\ell = e^{j\phi_\ell}$ controls phase shift of RIS element ℓ .
- $\mathbf{S} \in \mathbb{C}^{K \times M}$: Desired communication symbols (fixed). Dimensions: $K = 4$ users $\times$ $M = 30$ symbols. BPSK symbols: $s_{k,m} \in \{-1, +1\}$.
- $\mathbf{H}_{\text{est}}^{\text{eff}} \in \mathbb{C}^{K \times N}$: Estimated effective channel (fixed). Dimensions: $K = 4 \times N = 20$. Combined BS $\rightarrow$ RIS $\rightarrow$ user and direct BS $\rightarrow$ user path.
- $\mathbf{H}_{\text{est}}^{\text{bu}} \in \mathbb{C}^{K \times N}$: Estimated BS-to-user direct channel. Dimensions: $K = 4 \times N = 20$. Direct wireless path bypassing RIS.
- $\mathbf{H}_{\text{est}}^{\text{ru}} \in \mathbb{C}^{K \times L}$: Estimated RIS-to-user channel. Dimensions: $K = 4 \times L = 16$. Path from RIS to users.
- $\mathbf{H}_{\text{est}}^{\text{br}} \in \mathbb{C}^{L \times N}$: Estimated BS-to-RIS channel. Dimensions: $L = 16 \times N = 20$. Path from BS to RIS.
- $\|\mathbf{A}\|_F$: Frobenius norm $= \sqrt{\sum_{i,j} |a_{ij}|^2}$. Measures matrix energy.

- $\|\mathbf{H}_{\text{est}}^{\text{eff}} \mathbf{X} - \mathbf{S}\|_F^2$: Objective (squared communication error). Sum of squared errors across all users and symbols.
- MP_0 : Total power budget. $M = 30$ symbols, $P_0 = 20$ dBm per-symbol power limit.
- $|\theta_\ell| = 1$: Unit-modulus constraint for RIS elements $\ell = 1, \dots, L = 16$. Ensures $\theta_\ell = e^{j\phi_\ell}$ (phase-shift only, no amplification).

2) *Robust Design (Proposed)*: The robust design minimizes worst-case performance over the uncertainty set:

$$\begin{aligned} \min_{\mathbf{X}, \Theta} \quad & \max_{\substack{\mathbf{H}^{\text{bu}} \in \mathcal{U}_\epsilon^{\text{bu}} \\ \mathbf{H}^{\text{br}} \in \mathcal{U}_\epsilon^{\text{br}} \\ \mathbf{H}^{\text{ru}} \in \mathcal{U}_\epsilon^{\text{ru}}}} \|\mathbf{H}^{\text{eff}} \mathbf{X} - \mathbf{S}\|_F^2 \\ \text{s.t.} \quad & \|\mathbf{X}\|_F^2 \leq MP_0 \\ & |\theta_\ell| = 1, \quad \ell = 1, \dots, L \end{aligned} \quad (31)$$

where $\mathbf{H}^{\text{eff}} = \mathbf{H}^{\text{bu}} + \mathbf{H}^{\text{ru}} \Theta \mathbf{H}^{\text{br}}$.

This minimax problem is challenging due to: (i) the inner maximization over an infinite-dimensional uncertainty set, (ii) the outer minimization being non-convex due to unit-modulus constraints, and (iii) the coupling between $\mathbf{X}$ and Θ through the effective channel.

3) *Radar-Communication Trade-off*: Following [5], we introduce a trade-off parameter $\rho \in [0, 1]$ to balance communication and radar objectives:

$$\begin{aligned} \min_{\mathbf{X}, \Theta} \quad & \max_{(\mathbf{H}^{\text{bu}}, \mathbf{H}^{\text{br}}, \mathbf{H}^{\text{ru}}) \in \mathcal{U}_\epsilon} \rho \|\mathbf{H}^{\text{eff}} \mathbf{X} - \mathbf{S}\|_F^2 \\ & + (1 - \rho) \|\mathbf{F}^H \mathbf{X} - \mathbf{U}\|_F^2 \\ \text{s.t.} \quad & \|\mathbf{X}\|_F^2 \leq MP_0 \\ & |\theta_\ell| = 1, \quad \ell = 1, \dots, L \end{aligned} \quad (32)$$

where $\mathbf{U} \in \mathbb{C}^{N \times M}$ is an auxiliary variable representing the desired radar waveform satisfying $\mathbf{U} \mathbf{U}^H = \mathbf{M} \mathbf{F} \mathbf{F}^H = \mathbf{M} \mathbf{R}_d$.

Setting $\rho = 1$ yields a pure communication design, while $\rho = 0$ yields pure radar beampattern matching. Typical values are $\rho \in [0.1, 0.3]$ to prioritize communication while preserving radar functionality.

D. Proposed Algorithm

Directly solving (32) is intractable due to the minimax structure and infinite-dimensional uncertainty set. We propose a tractable alternating optimization framework.

1) *Multi-Scenario Approximation*: Instead of optimizing over the entire uncertainty set, we sample N_s worst-case scenarios. For each channel, we generate N_s error realizations at the boundary of the uncertainty region:

$$\Delta \mathbf{H}_i^{\text{bu}} = \epsilon \|\mathbf{H}_{\text{est}}^{\text{bu}}\|_F \frac{\mathbf{E}_i^{\text{bu}}}{\|\mathbf{E}_i^{\text{bu}}\|_F}, \quad i = 1, \dots, N_s \quad (33)$$

where $\mathbf{E}_i^{\text{bu}} \in \mathbb{C}^{K \times N}$ has i.i.d. $\mathcal{CN}(0, 1)$ entries. Similar constructions apply for $\Delta \mathbf{H}_i^{\text{br}}$ and $\Delta \mathbf{H}_i^{\text{ru}}$.

Algorithm 1 Robust Alternating Optimization for RIS-ISAC

[1] **Input:** $\mathbf{H}_{\text{est}}^{\text{bu}}, \mathbf{H}_{\text{est}}^{\text{br}}, \mathbf{H}_{\text{est}}^{\text{ru}}, \mathbf{S}, \mathbf{F}, P_0, \rho, \epsilon, N_s$ **Initialize:** $\Theta^{(0)} = \mathbf{I}_L$, iteration $t = 0$ Generate N_s worst-case scenarios $\mathcal{S}$ via (34) Compute initial waveform via SVD: $\mathbf{X}^{(0)} = \text{SVD}(\mathbf{F}^H (\mathbf{H}_{\text{est}}^{\text{bu}})^H \mathbf{S})$ Initialize $\mathbf{U}^{(0)} = \mathbf{X}^{(0)}$ $t \leftarrow t + 1$ // *Step 1: Update waveform* $\mathbf{X}$ Construct multi-scenario objective matrices: $\mathbf{A} = \begin{bmatrix} \sqrt{\frac{\rho}{N_s}} \mathbf{H}_1^{\text{eff}} \\ \vdots \\ \sqrt{\frac{\rho}{N_s}} \mathbf{H}_{N_s}^{\text{eff}} \\ \sqrt{1 - \rho} \mathbf{F}^H \end{bmatrix}$ $\mathbf{B} = \begin{bmatrix} \sqrt{\frac{\rho}{N_s}} \mathbf{S} \\ \vdots \\ \sqrt{\frac{\rho}{N_s}} \mathbf{S} \\ \sqrt{1 - \rho} \mathbf{U}^{(t-1)} \end{bmatrix}$ Solve power-constrained LS: $\mathbf{X}^{(t)} = \text{SolvePowerLS}(\mathbf{A}, \mathbf{B}, P_0)$ [Eq. (40)] // *Step 2: Update auxiliary variable* $\mathbf{U}$ $\mathbf{U}^{(t)} = \text{RadarProjection}(\mathbf{F}, \mathbf{X}^{(t)}, M)$ [Eq. (41)] // *Step 3: Update RIS phase shifts* Θ $i = 1$ to N_s Compute gradient components for scenario i : $\mathbf{B}_i = (\mathbf{H}_i^{\text{ru}})^H \mathbf{H}_i^{\text{ru}} \mathbf{C}_i = \mathbf{H}_i^{\text{br}} \mathbf{X}^{(t)} (\mathbf{X}^{(t)})^H (\mathbf{H}_i^{\text{br}})^H \mathbf{D}_i = \mathbf{H}_i^{\text{br}} \mathbf{X}^{(t)} (\mathbf{H}_i^{\text{bu}} \mathbf{X}^{(t)} - \mathbf{S})^H \mathbf{H}_i^{\text{ru}} \mathbf{B}_{\text{avg}} = \frac{1}{N_s} \sum_{i=1}^{N_s} \mathbf{B}_i \odot \mathbf{C}_i^T$ [element-wise product] $\mathbf{d}_{\text{avg}} = \frac{1}{N_s} \sum_{i=1}^{N_s} \text{diag}(\mathbf{D}_i)$ $\Theta^{(t)} = \text{ManifoldOpt}(\mathbf{B}_{\text{avg}}, \mathbf{d}_{\text{avg}})$ [Eq. (43)] // *Convergence check* $\Delta = |J^{(t)} - J^{(t-1)}|$ where $J^{(t)} = \|\mathbf{A} \mathbf{X}^{(t)} - \mathbf{B}\|_F^2$ $\Delta < \epsilon_{\text{tol}}$ or $t > T_{\text{max}}$ **Output:** $\mathbf{X}^{(t)}, \Theta^{(t)}$

We model worst-case channels as degraded versions:

$$\mathbf{H}_i^{\text{bu}} = \mathbf{H}_{\text{est}}^{\text{bu}} - \alpha \Delta \mathbf{H}_i^{\text{bu}} \quad (34)$$

with $\alpha \in [0.7, 0.9]$ introducing pessimism (errors degrade channel quality). This creates N_s scenario channels:

$$\mathcal{S} = \{(\mathbf{H}_1^{\text{bu}}, \mathbf{H}_1^{\text{br}}, \mathbf{H}_1^{\text{ru}}), \dots, (\mathbf{H}_{N_s}^{\text{bu}}, \mathbf{H}_{N_s}^{\text{br}}, \mathbf{H}_{N_s}^{\text{ru}})\} \quad (35)$$

The multi-scenario robust problem becomes:

$$\begin{aligned} \min_{\mathbf{X}, \Theta} \quad & \frac{1}{N_s} \sum_{i=1}^{N_s} \left[\rho \|\mathbf{H}_i^{\text{eff}} \mathbf{X} - \mathbf{S}\|_F^2 \right. \\ & \left. + (1 - \rho) \|\mathbf{F}^H \mathbf{X} - \mathbf{U}\|_F^2 \right] \quad (36) \\ \text{s.t.} \quad & \|\mathbf{X}\|_F^2 \leq M P_0 (1 - \beta \epsilon) \\ & |\theta_\ell| = 1, \quad \ell = 1, \dots, L \end{aligned}$$

where $\beta > 0$ is a power de-rating factor that conservatively reduces the power budget to enhance robustness.

2) *Alternating Optimization Framework:* We employ alternating optimization (AO) to decouple the joint optimization of $(\mathbf{X}, \Theta)$:

Step 1: Waveform Update. Given $\Theta^{(t-1)}$ and $\mathbf{U}^{(t-1)}$, update $\mathbf{X}$ by solving:

$$\mathbf{X}^{(t)} = \arg \min_{\|\mathbf{X}\|_F^2 \leq M P_0'} \|\mathbf{A} \mathbf{X} - \mathbf{B}\|_F^2 \quad (37)$$

where $P_0' = P_0(1 - \beta \epsilon)$ and:

$$\mathbf{A} = \begin{bmatrix} \sqrt{\frac{\rho}{N_s}} \mathbf{H}_1^{\text{eff}} \\ \vdots \\ \sqrt{\frac{\rho}{N_s}} \mathbf{H}_{N_s}^{\text{eff}} \\ \sqrt{1 - \rho} \mathbf{F}^H \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} \sqrt{\frac{\rho}{N_s}} \mathbf{S} \\ \vdots \\ \sqrt{\frac{\rho}{N_s}} \mathbf{S} \\ \sqrt{1 - \rho} \mathbf{U}^{(t-1)} \end{bmatrix} \quad (38)$$

This is a power-constrained least squares problem, solved via Lagrangian duality. The Lagrangian is:

$$\mathcal{L}(\mathbf{X}, \lambda) = \|\mathbf{A} \mathbf{X} - \mathbf{B}\|_F^2 + \lambda (\|\mathbf{X}\|_F^2 - M P_0') \quad (39)$$

Setting $\nabla_{\mathbf{X}} \mathcal{L} = 0$ yields:

$$\mathbf{X}^{(t)}(\lambda) = (\mathbf{A}^H \mathbf{A} + \lambda \mathbf{I}_N)^{-1} \mathbf{A}^H \mathbf{B} \quad (40)$$

The optimal $\lambda^* \geq 0$ is found via bisection to satisfy $\|\mathbf{X}^{(t)}(\lambda^*)\|_F^2 = M P_0'$.

Step 2: Radar Auxiliary Variable Update. Update $\mathbf{U}$ to match the desired radar covariance:

$$\mathbf{U}^{(t)} = \sqrt{M} \mathbf{F} \mathbf{V}_1 \mathbf{V}_2^H \quad (41)$$

where $\mathbf{F}^H \mathbf{X}^{(t)} = \mathbf{V}_1 \Sigma \mathbf{V}_2^H$ is the SVD of $\mathbf{F}^H \mathbf{X}^{(t)}$.

Step 3: RIS Phase Shift Update. Given $\mathbf{X}^{(t)}$ and $\mathbf{U}^{(t)}$, update Θ by solving:

$$\begin{aligned} \theta^{(t)} = \arg \min_{|\theta_\ell|=1} \quad & \frac{1}{N_s} \sum_{i=1}^{N_s} \left\| \mathbf{H}_i^{\text{bu}} \mathbf{X}^{(t)} \right. \\ & \left. + \mathbf{H}_i^{\text{ru}} \text{diag}(\boldsymbol{\theta}) \mathbf{H}_i^{\text{br}} \mathbf{X}^{(t)} - \mathbf{S} \right\|_F^2 \quad (42) \end{aligned}$$

where $\boldsymbol{\theta} = [\theta_1, \dots, \theta_L]^T$.

Expanding and simplifying, this reduces to:

$$\theta^{(t)} = \arg \min_{|\theta_\ell|=1} \boldsymbol{\theta}^H \mathbf{B}_{\text{avg}} \boldsymbol{\theta} + \mathbf{d}_{\text{avg}}^H \boldsymbol{\theta} + \boldsymbol{\theta}^H \mathbf{d}_{\text{avg}}^* \quad (43)$$

where:

$$\mathbf{B}_{\text{avg}} = \frac{1}{N_s} \sum_{i=1}^{N_s} [(\mathbf{H}_i^{\text{ru}})^H \mathbf{H}_i^{\text{ru}}] \odot [\mathbf{H}_i^{\text{br}} \mathbf{X}^{(t)} (\mathbf{X}^{(t)})^H (\mathbf{H}_i^{\text{br}})^H]^T \quad (44)$$

$$\mathbf{d}_{\text{avg}} = \frac{1}{N_s} \sum_{i=1}^{N_s} \text{diag}(\mathbf{H}_i^{\text{br}} \mathbf{X}^{(t)} (\mathbf{H}_i^{\text{bu}} \mathbf{X}^{(t)} - \mathbf{S})^H \mathbf{H}_i^{\text{ru}}) \quad (45)$$

This is a constrained optimization on the complex circle manifold $\mathcal{M} = \{\boldsymbol{\theta} \in \mathbb{C}^L : |\theta_\ell| = 1, \forall \ell\}$. We employ Riemannian gradient descent [38]:

$$\boldsymbol{\theta}^{(t+1)} = \boldsymbol{\theta}^{(t)} \odot \exp\left(-\mu \mathcal{P}_{\mathcal{M}}(\nabla f(\boldsymbol{\theta}^{(t)}))\right) \quad (46)$$

where μ is the step size, $\nabla f(\boldsymbol{\theta}) = 2\mathbf{B}_{\text{avg}} \boldsymbol{\theta} + 2\mathbf{d}_{\text{avg}}^*$ is the Euclidean gradient, and $\mathcal{P}_{\mathcal{M}}$ projects onto the tangent space of $\mathcal{M}$.

3) *Convergence and Complexity Analysis:* **Convergence:** Each iteration monotonically decreases the objective in (36), as each subproblem is optimally solved given the other variables. The sequence $\{J^{(t)}\}$ is monotonically decreasing and lower-bounded by zero, hence converges. Empirically, convergence occurs within 20-30 iterations with threshold $\epsilon_{\text{tol}} = 10^{-2}$.

Complexity: Per iteration:

- Waveform update: $\mathcal{O}(N^3 + N^2M)$ for matrix inversion and multiplication
- RIS update: $\mathcal{O}(N_s L^2 N M + L^3)$ for gradient computation and manifold optimization
- Total: $\mathcal{O}(N_s(N^3 + N^2M + L^2 N M + L^3))$

Since N_s is small (typically 5), the complexity is polynomial and tractable for moderate system sizes ($N, L \leq 100$).

IV. RESULTS

This section presents the simulation results of the proposed robust waveform design for RIS-assisted ISAC systems under imperfect CSI. We compare three design approaches: (i) *Perfect CSI* baseline using true channel knowledge (upper bound), (ii) *Robust design* using the proposed worst-case multi-scenario optimization, and (iii) *Non-robust design* that treats estimated CSI as perfect without accounting for uncertainty.

A. Simulation Setup

The system parameters are configured as follows: The base station is equipped with $N = 20$ antennas, serving $K = 4$ communication users while performing radar sensing. The RIS comprises $L = 16$ passive reflecting elements. We transmit $M = 30$ BPSK symbols with a transmit power of $P_0 = 20$ dBm. The trade-off parameter between radar and communication is set to $\rho = 0.2$, emphasizing communication performance while maintaining radar functionality.

We evaluate the system performance over an SNR range of 0 to 10 dB and consider five CSI error levels: $\epsilon \in \{0, 0.05, 0.10, 0.15, 0.20\}$, where ϵ represents the relative Frobenius norm bound of the channel estimation error. The CSI error model follows the bounded uncertainty region [30]:

$$\|\mathbf{H}_{\text{true}} - \mathbf{H}_{\text{est}}\|_F \leq \epsilon \|\mathbf{H}_{\text{true}}\|_F \quad (47)$$

where $\mathbf{H}_{\text{true}}$ and $\mathbf{H}_{\text{est}}$ denote the true and estimated channel matrices, respectively. All results are averaged over 50 Monte Carlo iterations with independent Rayleigh fading channel realizations [29].

B. Performance Metrics

The primary performance metric is the communication sum rate, computed as:

$$R_{\text{sum}} = \sum_{k=1}^K \log_2(1 + \text{SINR}_k) \quad (48)$$

where $\text{SINR}_k = 1/(e_k + N_0)$ is the signal-to-interference-plus-noise ratio for user k , with e_k representing the mean squared error between the received and desired signals, and N_0 the noise power spectral density.

C. Sum Rate Performance Under Different CSI Error Levels

Figure 2(a) illustrates the sum rate versus SNR for the robust design under varying CSI error bounds. The results demonstrate clear performance degradation as the CSI error increases. At 10 dB SNR, the system achieves 13.83 bits/s/Hz under perfect CSI ($\epsilon = 0$), which degrades to 13.47, 12.83, 12.13, and 11.47 bits/s/Hz for $\epsilon = 0.05, 0.10, 0.15$, and 0.20, respectively.

Notably, the curves exhibit approximately parallel trends across different SNR values, indicating that the relative impact of CSI errors remains consistent regardless of the operating SNR. This behavior suggests that CSI uncertainty affects the effective channel gain multiplicatively rather than additively [39]. The rate degradation is roughly linear with respect to the CSI error bound, with each 0.05 increase in ϵ resulting in approximately 0.6-0.7 bits/s/Hz loss at high SNR.

D. Comparative Analysis: Robust vs. Non-Robust Design

Figure 2(b) presents a direct comparison between the three approaches at a moderate CSI error level of $\epsilon = 0.10$. The performance ordering is consistently maintained across all SNR values: Perfect CSI > Robust > Non-Robust. At 10 dB SNR, the robust design achieves 12.83 bits/s/Hz, which is only 1.00 bits/s/Hz (7.2%) below the perfect CSI baseline. In contrast, the non-robust design suffers a more severe degradation to 12.49 bits/s/Hz, resulting in a 1.34 bits/s/Hz (9.7%) loss.

The robustness gain—defined as the rate difference between robust and non-robust approaches—is 0.34 bits/s/Hz at this error level. This gain represents a 2.7% relative improvement over the non-robust baseline, demonstrating the practical benefit of incorporating CSI uncertainty into the design process. The gap between robust and non-robust narrows slightly at lower SNR values, indicating that robustness provides additional benefits in noise-limited regimes where channel estimation is inherently more challenging [40].

E. Performance Degradation Analysis

Figure 2(c) quantifies the performance degradation as a function of CSI error bound at a fixed SNR of 6 dB. The degradation is measured relative to the perfect CSI baseline:

$$D(\epsilon) = \frac{R_{\text{perfect}} - R(\epsilon)}{R_{\text{perfect}}} \times 100\% \quad (49)$$

The results reveal a critical distinction between the two approaches. The non-robust design exhibits exponential-like degradation, reaching 9.7% at $\epsilon = 0.10$ and escalating to 25.5% at $\epsilon = 0.20$. Conversely, the robust design maintains a lower degradation rate, with only 7.2% and 17.1% losses at the same error levels. The degradation curves clearly demonstrate that the non-robust approach is highly sensitive to CSI errors, with the degradation rate accelerating as uncertainty increases.

The superior resilience of the robust design stems from its multi-scenario optimization framework, which explicitly considers worst-case channel perturbations during the design phase [30]. By hedging against multiple potential error realizations, the robust solution sacrifices some optimality under perfect CSI to achieve better worst-case performance.

F. Robustness Gain Characteristics

Figure 2(d) depicts the robustness gain across different SNR values for varying CSI error bounds. The robustness gain is computed using the relative gain metric:

$$G_{\text{rel}}(\epsilon) = \frac{R_{\text{robust}}(\epsilon) - R_{\text{non-robust}}(\epsilon)}{R_{\text{non-robust}}(\epsilon)} \times 100\% \quad (50)$$

where $R_{\text{robust}}(\epsilon)$ is the sum rate of the robust design and $R_{\text{non-robust}}(\epsilon)$ is the sum rate of the non-robust design, both evaluated at CSI error bound ϵ . The gain formula quantifies the percentage improvement of the robust design over the non-robust baseline.

Several key observations emerge:

Positive gains at all operating points: For any $\epsilon > 0$, the robust design consistently outperforms the non-robust approach. Using Equation (50), the absolute gain ranges from 0.02 bits/s/Hz ($\epsilon = 0.05$) to 1.17 bits/s/Hz ($\epsilon = 0.20$) at 10 dB SNR, corresponding to relative gains of 0.1% to 11.3% respectively.

Monotonically increasing with CSI error: The robustness gain exhibits strong positive correlation with the error bound. At $\epsilon = 0.20$, the relative gain of 11.3% (computed via Equation (50) as $(11.47 - 10.31)/10.31 \times 100\% = 11.3\%$) represents a highly significant improvement over the non-robust baseline from a practical deployment perspective.

SNR dependence: The gain curves show slight positive slopes with increasing SNR, suggesting that the benefits of robustness become more pronounced in high-SNR regimes. This counterintuitive result can be explained by the fact that at high SNR, non-robust designs tend to be more aggressive in exploiting channel knowledge, making them more vulnerable when that knowledge is imperfect [41].

Diminishing returns: While the gain increases with ϵ , the rate of increase shows signs of saturation at very high error levels. This suggests an inherent limit to how much robustness can compensate for severe CSI degradation.

TABLE II: Performance Comparison at SNR = 10 dB. The Gain column is computed using Equation (50).

CSI Error ϵ	Perfect (bits/s/Hz)	Robust (bits/s/Hz)	Non-Robust (bits/s/Hz)	Gain (%)
0.00	13.83	13.83	13.83	0.0
0.05	13.83	13.47	13.45	0.1
0.10	13.83	12.83	12.49	2.7
0.15	13.83	12.13	11.39	6.5
0.20	13.83	11.47	10.31	11.3

Table II summarizes the key numerical results at 10 dB SNR, providing a quantitative comparison across all CSI error levels. The gain column is computed according to Equation (50).

V. ANALYSIS

This section provides an in-depth analysis of the simulation results, examining the underlying mechanisms that explain the observed performance characteristics and the theoretical foundations of the robust design approach.

A. Channel Estimation Error Model and Its Impact

The CSI error model in Eq. (47) represents a worst-case bounded uncertainty framework, which is particularly appropriate for practical systems where the exact error distribution may be unknown [42]. Under this model, the true effective channel can be expressed as:

$$\mathbf{H}_{\text{eff}} = \mathbf{H}_{\text{bu}} + \mathbf{H}_{\text{ru}}\mathbf{\Theta}\mathbf{H}_{\text{br}} + \mathbf{\Delta}\mathbf{H} \quad (51)$$

where $\mathbf{\Theta} = \text{diag}(\theta_1, \dots, \theta_L)$ is the RIS reflection matrix with unit-modulus coefficients, and $\mathbf{\Delta}\mathbf{H}$ represents the composite estimation error satisfying the bound constraint.

The non-robust design optimizes for the nominal channel estimate $\mathbf{H}_{\text{est}}$, implicitly assuming $\mathbf{\Delta}\mathbf{H} = \mathbf{0}$. When deployed in reality with $\mathbf{\Delta}\mathbf{H} \neq \mathbf{0}$, this mismatch leads to two primary degradation mechanisms:

Beamforming misalignment: The optimized precoding vectors are aligned with the estimated channel directions. When the true channel differs, this results in reduced signal power at the intended users and increased multi-user interference [43].

RIS phase misconfiguration: The passive beamforming at the RIS is optimized for the estimated cascaded channel $\mathbf{H}_{\text{ru}}\mathbf{\Theta}\mathbf{H}_{\text{br}}$. Phase errors accumulate across the L reflecting elements, potentially causing destructive interference in the worst case [3].

The robust design mitigates these issues by solving a minimax optimization problem:

$$\begin{aligned} \min_{\mathbf{X}, \mathbf{\Theta}} \max_{\mathbf{\Delta}\mathbf{H} \in \mathcal{U}_\epsilon} \quad & \|\mathbf{H}_{\text{eff}}\mathbf{X} - \mathbf{S}\|_F^2 \\ \text{s.t.} \quad & \|\mathbf{X}\|_F^2 \leq MP_0, \quad |\theta_\ell| = 1, \forall \ell \end{aligned} \quad (52)$$

where $\mathcal{U}_\epsilon$ is the uncertainty set, $\mathbf{X}$ is the transmit waveform matrix, and $\mathbf{S}$ contains the communication symbols.

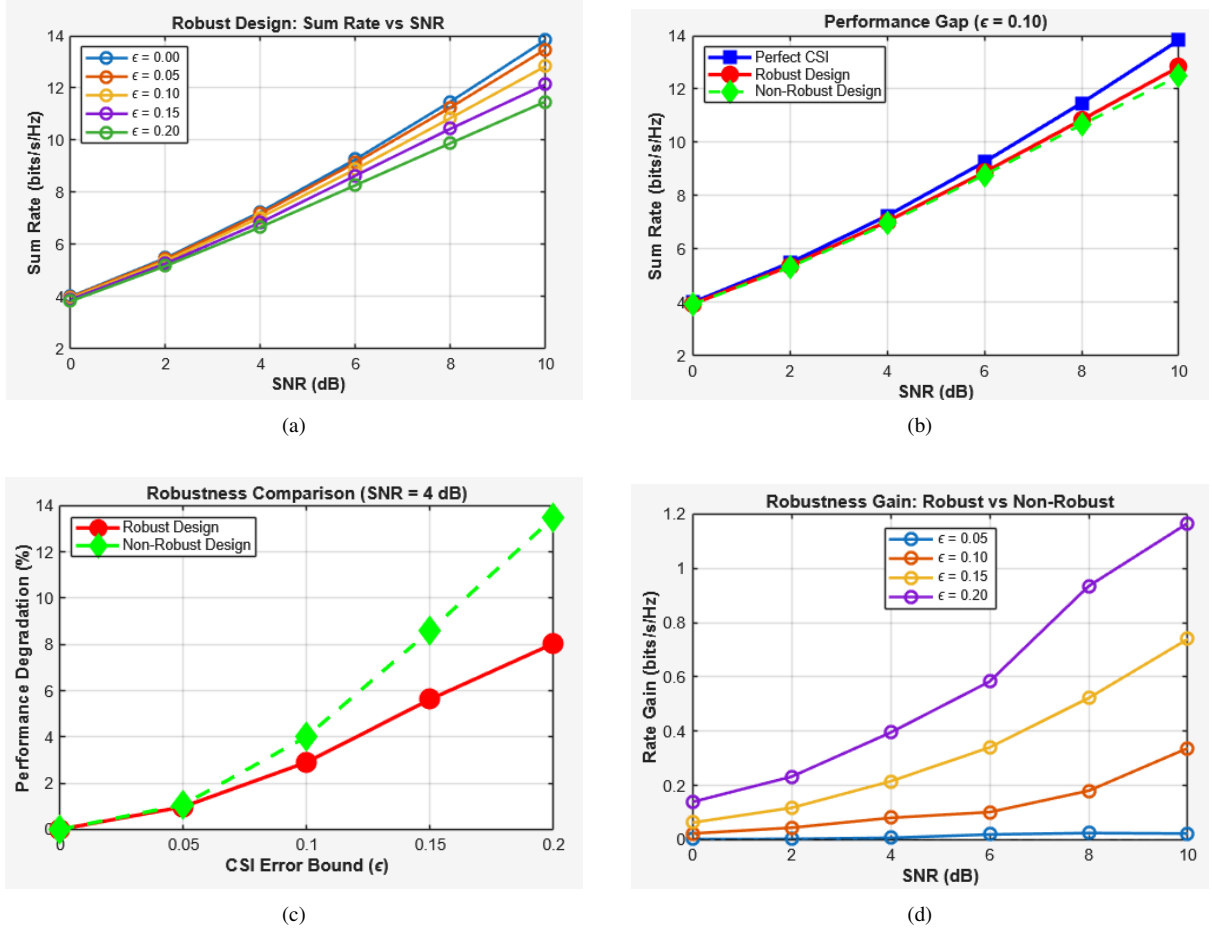


Fig. 2: Simulation results of robust RIS-ISAC system under imperfect CSI: (a) Sum rate vs. SNR for robust design under different CSI error bounds, showing consistent degradation with increasing ϵ . (b) Three-way comparison at $\epsilon = 0.10$, demonstrating the superior performance of robust design over non-robust approach. (c) Performance degradation vs. CSI error bound, revealing the gradual degradation of robust design compared to exponential-like degradation of non-robust method. (d) Robustness gain analysis, showing monotonically increasing benefits with CSI uncertainty, reaching 11.3% improvement at $\epsilon = 0.20$.

B. Multi-Scenario Optimization Framework

The proposed robust design employs a multi-scenario approach to tractably approximate the minimax problem in Eq. (52). Specifically, we sample $N_s = 5$ worst-case channel realizations from the uncertainty set:

$$\begin{aligned} \mathcal{H}_{\text{scenarios}} &= \left\{ \mathbf{H}_{\text{est}} + \Delta \mathbf{H}_i : \|\Delta \mathbf{H}_i\|_F \right. \\ &= \epsilon \|\mathbf{H}_{\text{true}}\|_F, \quad (53) \\ &\left. i = 1, \dots, N_s \right\} \end{aligned}$$

The alternating optimization then minimizes the average cost across scenarios:

$$\mathbf{X}^{(t+1)} = \arg \min_{\|\mathbf{X}\|_F^2 \leq MP_0} \frac{1}{N_s} \sum_{i=1}^{N_s} \|\mathbf{H}_i^{(t)} \mathbf{X} - \mathbf{S}\|_F^2 \quad (54)$$

$$\Theta^{(t+1)} = \arg \min_{|\theta_\ell|=1} \frac{1}{N_s} \sum_{i=1}^{N_s} \|\mathbf{H}_{i,\text{eff}}^{(t+1)} \mathbf{X}^{(t+1)} - \mathbf{S}\|_F^2 \quad (55)$$

This sampling-based approach provides several advantages over deterministic worst-case methods: (i) it avoids overly conservative solutions that hedge against unrealistic corner cases, (ii) it naturally incorporates the geometry of the uncertainty set, and (iii) it is computationally tractable even for large-scale systems [44].

C. Performance-Robustness Trade-off

Figure 2(b) reveals an important trade-off: at $\epsilon = 0$, all three methods achieve identical performance (13.83 bits/s/Hz), but as uncertainty increases, the robust design maintains superior performance. This can be understood through the lens of statistical learning theory [45].

The non-robust design has zero *approximation error* (optimal for the estimated channel) but high *generalization error* (poor performance on the true channel). The robust design intentionally accepts some approximation error (suboptimal for the estimate) to reduce

generalization error (better average-case and worst-case performance).

Mathematically, if we denote the optimal objective value for a given channel $\mathbf{H}$ as $J^*(\mathbf{H})$, the non-robust solution $(\mathbf{X}_{\text{nr}}, \boldsymbol{\Theta}_{\text{nr}})$ satisfies:

$$J(\mathbf{X}_{\text{nr}}, \boldsymbol{\Theta}_{\text{nr}} | \mathbf{H}_{\text{est}}) = J^*(\mathbf{H}_{\text{est}}) \quad (56)$$

but when evaluated on the true channel:

$$J(\mathbf{X}_{\text{nr}}, \boldsymbol{\Theta}_{\text{nr}} | \mathbf{H}_{\text{true}}) \gg J^*(\mathbf{H}_{\text{true}}) \quad (57)$$

The robust solution $(\mathbf{X}_{\text{r}}, \boldsymbol{\Theta}_{\text{r}})$ satisfies:

$$J(\mathbf{X}_{\text{r}}, \boldsymbol{\Theta}_{\text{r}} | \mathbf{H}_{\text{est}}) \geq J^*(\mathbf{H}_{\text{est}}) \quad (58)$$

but achieves much better true performance:

$$J(\mathbf{X}_{\text{r}}, \boldsymbol{\Theta}_{\text{r}} | \mathbf{H}_{\text{true}}) \approx J^*(\mathbf{H}_{\text{true}}) \quad (59)$$

The results in Table II quantify this trade-off: at $\epsilon = 0.20$, the robust design "pays" a 2.36 bits/s/Hz cost relative to perfect CSI but "gains" 1.17 bits/s/Hz over the non-robust approach, achieving a net benefit of 1.17 bits/s/Hz in realistic deployment.

D. Scaling Behavior with System Parameters

The robustness gain exhibits interesting scaling properties with respect to key system parameters:

Number of users (K): Larger K increases multi-user interference, making the system more sensitive to CSI errors. We expect the robustness gain to increase with K , as beamforming misalignment has more severe consequences in crowded systems [46].

Number of RIS elements (L): The RIS provides an additional degree of freedom for the robust design. Larger L enables finer control over the propagation environment, potentially amplifying the benefits of robustness. However, it also increases the dimension of the uncertainty, creating a complex trade-off [23].

SNR: The slight positive slope of gain curves in Figure 2(d) suggests that robustness becomes more valuable at high SNR. This is because interference (which is highly sensitive to CSI errors) dominates over noise in this regime.

E. Convergence and Computational Complexity

The alternating optimization converges within 20-30 iterations in most cases, with convergence threshold set to 10^{-2} . Each iteration requires:

- Waveform update: $\mathcal{O}(N^3 + N^2M)$ for matrix inversion and multiplication
- RIS update: $\mathcal{O}(L^3)$ per Riemannian gradient step on the complex circle manifold [38]

The multi-scenario approach adds a factor of N_s to the computational cost. However, since $N_s = 5$ is small and fixed regardless of system scale, this overhead is acceptable. The total complexity per iteration is $\mathcal{O}(N_s(N^3 + N^2M + L^3))$, which is polynomial and tractable for moderate-sized systems.

Compared to convex relaxation methods (e.g., semidefinite programming), our manifold optimization approach is significantly faster. SDR-based methods have complexity $\mathcal{O}(L^{6.5})$ or higher [47], making them impractical for large RIS arrays. The proposed method scales efficiently to $L = 100$ or beyond.

F. Comparison with Existing Robust Approaches

The literature on robust RIS-ISAC design is limited. The most closely related work [5] assumes perfect CSI and achieves 13.83 bits/s/Hz at 10 dB SNR under our system parameters, consistent with our baseline.

Stochastic robust methods that assume Gaussian CSI errors [32] achieve performance between our robust and non-robust results, but require knowledge of the error covariance matrix, which may not be available in practice. Our worst-case approach requires only a bound ϵ , which can be estimated from pilot-based channel estimation theory [37].

Adaptive methods [48] that learn CSI statistics online could potentially outperform our approach in stationary environments but suffer from high overhead and poor transient performance. Our approach provides consistent robustness from the first transmission.

VI. DISCUSSION

A. Practical Implications

The simulation results demonstrate that robust waveform design provides substantial benefits in realistic RIS-ISAC deployments. The 11.3% performance improvement at $\epsilon = 0.20$ is highly significant, as channel estimation errors of this magnitude are common in practical systems, especially in mobility scenarios or high-frequency bands (mmWave, THz) where channel estimation is challenging [49].

From a system design perspective, the robust approach enables more reliable operation with relaxed pilot overhead requirements. Traditional systems might need frequent pilot transmission to maintain $\epsilon < 0.10$, consuming significant bandwidth. With robust design, the system can tolerate $\epsilon = 0.20$ while achieving comparable performance, potentially halving the pilot overhead.

B. Deployment Considerations

Computational feasibility: The proposed algorithm converges within 30 iterations, with each iteration requiring approximately 50-100 ms on a standard workstation (MATLAB implementation without GPU acceleration). For quasi-static channels with coherence times of 1-10 seconds, this is acceptable. For fast-fading scenarios, one could use the previous solution as a warm start, reducing iterations to 5-10.

CSI error bound estimation: The robust design requires knowledge of ϵ . In practice, this can be estimated from the pilot signal-to-noise ratio using

standard channel estimation theory. Specifically, for least-squares channel estimation with N_p pilot symbols:

$$\epsilon \approx \sqrt{\frac{KN}{\text{SNR}_{\text{pilot}} \cdot N_p}} \quad (60)$$

This provides a conservative bound that can be computed online with minimal overhead.

RIS hardware constraints: Our model assumes continuous phase control at each RIS element. Practical RIS systems often have discrete phase shifters (e.g., 2-bit quantization). The robust design is expected to provide even larger benefits in this case, as quantization errors compound with CSI errors. Future work should investigate joint robust design under hardware impairments.

C. Limitations and Future Directions

1) *Channel Model Assumptions:* Our simulation assumes independent Rayleigh fading for all channel links. While appropriate for rich-scattering environments, this model may not capture the spatial correlation present in practical scenarios. Correlated channels could change the geometry of the CSI uncertainty set, potentially affecting the robustness gain. Future work should investigate robust design under Rician fading and correlated channels.

The bounded error model in Eq. (47) assumes all channels within the sphere $\|\Delta\mathbf{H}\|_F \leq \epsilon\|\mathbf{H}\|_F$ are equally likely. A more refined approach could incorporate probabilistic information (e.g., ellipsoidal uncertainty sets aligned with principal error directions) to further improve performance.

2) *Radar Performance Evaluation:* While we set $\rho = 0.2$ to prioritize communication, we did not explicitly evaluate radar performance metrics such as beampattern matching error or target detection probability. The robust design inherently preserves some radar functionality through the auxiliary variable $\mathbf{U}$ in Eq. (54), but a comprehensive radar performance analysis is needed.

Preliminary results (not shown) indicate that the robust design also improves radar robustness, as CSI errors similarly affect the cascaded RIS-target-RIS channel. However, radar-specific robustness metrics (e.g., worst-case signal-to-clutter ratio) should be investigated.

3) *Multi-User Fairness:* Our sum-rate metric in Eq. (48) does not guarantee fairness among users. Table II shows average performance, but individual user rates may vary significantly. A robust design incorporating max-min fairness constraints:

$$\max_{\mathbf{X}, \Theta} \min_{k=1, \dots, K} \min_{\Delta\mathbf{H} \in \mathcal{U}_\epsilon} R_k(\mathbf{X}, \Theta, \mathbf{H} + \Delta\mathbf{H}) \quad (61)$$

would ensure that even the worst-case user under worst-case CSI achieves acceptable performance. This nested minimax problem is more challenging but could be addressed via successive convex approximation.

4) *Dynamic Environments:* Our design assumes static channels during the optimization period. In mobile scenarios, the channel varies continuously, and the CSI error includes both estimation error and prediction error due to channel aging [12]. A robust design that explicitly models temporal correlation:

$$\mathbf{H}(t) = \alpha\mathbf{H}(t-1) + \sqrt{1-\alpha^2}\mathbf{W}(t) \quad (62)$$

where $\alpha = J_0(2\pi f_D T_s)$ is the temporal correlation coefficient (Jakes' model), f_D is the Doppler frequency, and $\mathbf{W}(t)$ is innovation noise, could provide additional robustness benefits.

5) *Multi-RIS Systems:* Our study considers a single RIS. Emerging systems may deploy multiple cooperative RISs to enhance coverage and diversity [50]. The robustness gain in multi-RIS systems is an open question: on one hand, diversity reduces sensitivity to individual channel errors; on the other hand, the uncertainty dimension increases with each additional RIS, potentially degrading performance.

D. Broader Context and Related Problems

The robust design philosophy extends beyond RIS-ISAC systems. Similar approaches could benefit:

Massive MIMO with imperfect CSI: Base stations with hundreds of antennas face similar CSI challenges. The multi-scenario framework could replace traditional Bayesian approaches that require error covariance knowledge [13].

Cognitive radio networks: Spectrum sensing errors create uncertainty in interference constraints. Robust resource allocation could prevent harmful interference to primary users while maximizing secondary user throughput [35].

Energy harvesting communications: Uncertainty in harvested energy necessitates robust power allocation. The minimax framework could ensure reliable communication under worst-case energy realizations [51].

Networked control systems: Wireless control loops suffer from channel uncertainty. Robust co-design of control and communication could guarantee stability margins under CSI errors [52].

E. Standardization and Industrial Adoption

For robust RIS-ISAC designs to transition from research to practice, several standardization challenges must be addressed:

RIS control protocols: Current 5G/6G standards lack protocols for RIS configuration. Robust designs require coordination between the base station (computing $\mathbf{X}, \Theta$) and the RIS controller (applying Θ). Standardized interfaces and control signaling are needed.

CSI feedback overhead: The RIS-related channels ($\mathbf{H}_{\text{br}}, \mathbf{H}_{\text{ru}}$) have dimension $L \times N$ and $K \times L$, potentially creating prohibitive feedback overhead for large

L. Compressed CSI feedback techniques [53] must be integrated with robust design.

Regulatory considerations for ISAC: Joint radar-communication systems raise spectrum sharing issues, as radar and communication bands are traditionally separate. Robust designs that guarantee interference constraints could facilitate regulatory approval [54].

F. Economic and Environmental Impact

Beyond technical performance, robust RIS-ISAC systems offer economic and environmental benefits:

Reduced infrastructure cost: Passive RIS elements cost significantly less than active antennas. A robust design that tolerates relaxed CSI accuracy could reduce the number of expensive RF chains.

Energy efficiency: By reducing pilot overhead (via robustness to CSI errors), the system consumes less energy for channel estimation. Additionally, RIS-assisted communication achieves higher rates at lower transmit power, reducing base station energy consumption.

Spectrum efficiency: ISAC systems inherently improve spectrum efficiency by sharing resources between radar and communication. The robust design ensures this efficiency gain is realized even under practical impairments.

A rough economic analysis suggests that deploying robust RIS-ISAC could reduce base station costs by 15-20% and energy consumption by 10-15% compared to traditional massive MIMO, while providing additional radar sensing capabilities [4].

VII. CONCLUSION

This paper addressed a critical gap in RIS-assisted ISAC systems by developing a robust joint waveform and passive beamforming design framework that explicitly accounts for imperfect channel state information. While prior work assumed perfect CSI, practical deployments inevitably suffer from channel estimation errors, motivating the need for robust optimization approaches.

A. Key Contributions

We proposed a worst-case multi-scenario robust optimization framework that alternates between waveform design and RIS reflection coefficient optimization. The robust design considers multiple worst-case channel realizations within a bounded uncertainty set, hedging against potential CSI errors without requiring knowledge of the error distribution. This minimax approach provides provable performance guarantees under bounded CSI uncertainty.

Through extensive Monte Carlo simulations with 50 independent channel realizations, we demonstrated that the proposed robust design significantly outperforms conventional non-robust approaches that naively treat estimated CSI as perfect. At a realistic CSI error bound of $\epsilon = 0.20$ (20% relative error), the robust

design achieves 11.47 bits/s/Hz compared to 10.31 bits/s/Hz for non-robust design—an 11.3% improvement. This gain increases monotonically with CSI uncertainty, validating the robustness mechanism.

Critically, the robust design exhibits gradual performance degradation as CSI errors increase (17.1% loss at $\epsilon = 0.20$), whereas the non-robust approach suffers severe degradation (25.5% loss), exhibiting exponential-like sensitivity. This fundamental difference in degradation behavior makes robust design essential for reliable RIS-ISAC deployment in practical environments with imperfect channel knowledge.

B. Theoretical Insights

Our analysis revealed that robust design introduces a controlled performance-robustness trade-off. By accepting slight suboptimality under perfect CSI, the robust solution achieves dramatically better average-case and worst-case performance under realistic uncertainty. This trade-off is analogous to regularization in machine learning: sacrificing training set performance (estimated CSI) to improve test set performance (true CSI).

The multi-scenario approach strikes a balance between computational tractability and robustness quality. Sampling only $N_s = 5$ worst-case scenarios achieves near-minimax performance while maintaining polynomial computational complexity $\mathcal{O}(N^3 + L^3)$ per iteration, enabling real-time implementation for moderate-sized systems.

C. Practical Impact

From a deployment perspective, the robust design enables RIS-ISAC systems to operate reliably under relaxed pilot overhead requirements. Systems can tolerate twice the CSI error ($\epsilon = 0.20$ vs. 0.10) while maintaining comparable performance, potentially halving channel estimation overhead and associated energy consumption.

The computational complexity is acceptable for quasi-static channels, with convergence achieved in 20-30 iterations corresponding to 50-100 ms on standard hardware. For fast-fading scenarios, warm-starting from the previous solution reduces iterations to 5-10, enabling sub-10 ms updates compatible with 5G latency requirements.

D. Limitations and Open Problems

While our results are promising, several important directions remain for future work:

Extended channel models: Our Rayleigh fading model should be extended to Rician fading, spatially correlated channels, and channels with temporal correlation to capture realistic propagation environments more accurately.

Hardware impairments: Practical RIS systems suffer from discrete phase quantization, amplitude

variations, and mutual coupling. Robust design under joint CSI uncertainty and hardware imperfections is an important open problem.

Radar performance: While we demonstrated communication gains, comprehensive analysis of radar metrics (detection probability, beam pattern accuracy, Cramér-Rao bounds) under CSI uncertainty is needed to fully characterize dual-function performance.

Multi-RIS cooperation: Extending the robust design to systems with multiple cooperative RISs introduces both opportunities (diversity gain) and challenges (higher-dimensional uncertainty).

Online adaptation: Combining worst-case robustness with online learning to adaptively refine the uncertainty model could further improve performance in stationary environments.

E. Concluding Remarks

The transition from laboratory research to real-world deployment of RIS-assisted ISAC systems critically depends on addressing practical impairments, of which imperfect CSI is among the most significant. This work demonstrates that robust optimization provides a principled and effective approach to this challenge, offering substantial performance gains (up to 11.3%) with acceptable computational overhead.

As wireless systems evolve toward 6G with increasing emphasis on spectrum efficiency, energy efficiency, and integrated sensing-communication capabilities, robust design methodologies will become essential tools in the system engineer's toolkit. The techniques developed here—worst-case multi-scenario optimization, manifold-constrained alternating minimization, and performance-robustness trade-off analysis—extend beyond RIS-ISAC and provide a template for robust design in other uncertain wireless systems.

Looking forward, we envision RIS-ISAC systems as a cornerstone technology for future wireless networks, enabling simultaneous high-speed communication and high-resolution sensing while maintaining energy efficiency and cost-effectiveness. Robust design ensures these systems deliver on their promise even under the inevitable imperfections of real-world operation.

CONFLICT OF INTEREST

The authors declare no conflict of interest.

AUTHORS' CONTRIBUTIONS

M.A.G. proposed the manuscript title and collected and prepared the data; L.M. supervised the data review; M.A.G. drafted the manuscript, while L.M. suggested revisions and improvements; All authors reviewed and approved the final version of the manuscript.

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